UAI.PY: FACILITATING ACCESS TO INTERNET OF THINGS SOLUTIONS

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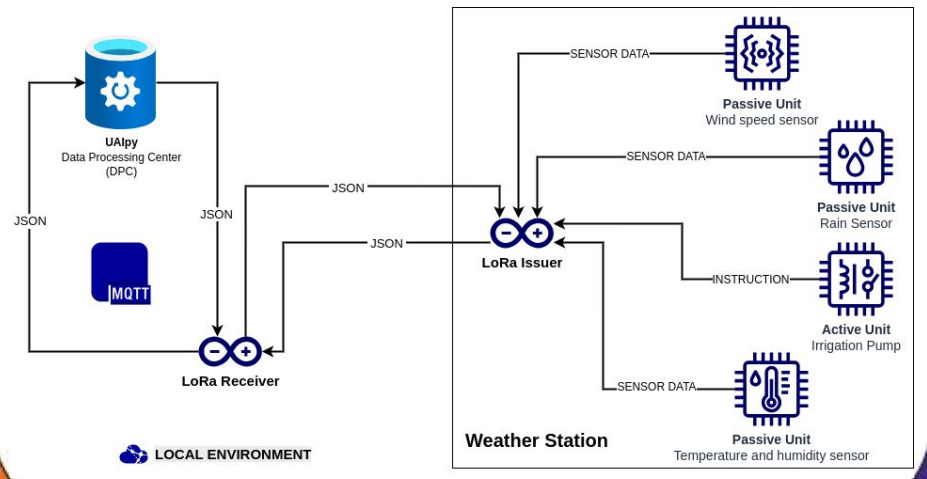
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INTRODUCTION

According to projections by the Food and Agriculture Organization of the United Nations (FAO, 2023), population growth is expected to be significant by 2050, with an increase of 1.15 billion people. With regard to agriculture, for example, this growth represents a gap of approximately 56% between food production in 2010 and projected needs in the future (World Resources Institute, 2019). In light of this, we designed a data processing center using seized TXBoxes. Adapted for IoT systems, these devices work to communicate and manage agricultural data, bringing greater efficiency and productivity. By reusing electronics, we hope to have a positive impact on productivity at a low environmental cost.

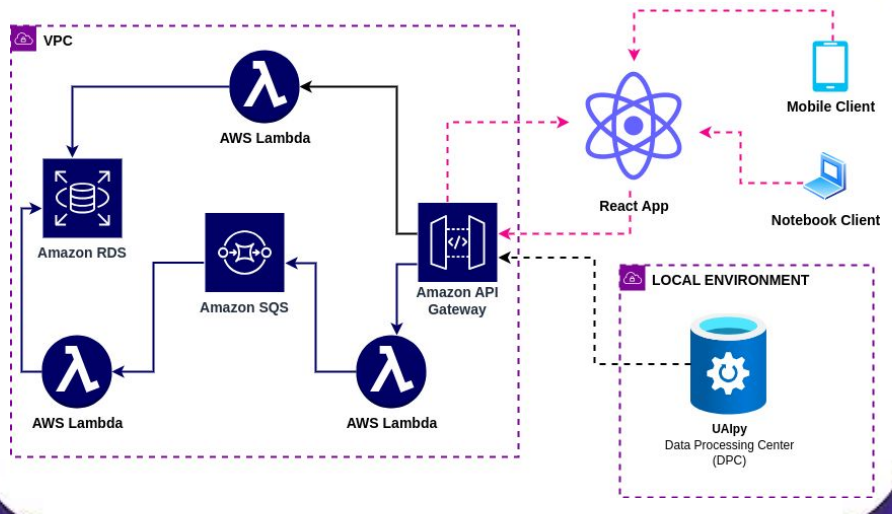
THE ROBOT (LOCAL ENVIRONMENT)



OBJECTIVES

This work is based on the combination of sustainable and low-cost technologies to manage data in distributed environments, resulting in an IoT solution that enhances the usability and scalability of a device called by UAI.py. By recycling a TVBox and replacing its software with the Armbian distribution (based on Linux), algorithms were developed to receive and send data to a web server using cloud services such as Amazon Web Services (AWS). This data is measured locally by sensors connected to the ESP32-LoRa module, which allows long-distance wireless communication. The operation uses the MQTT protocol, ensuring efficient and reliable transmission, even in places with limited connectivity.

RESULTS AND CONCLUSIONS



MATERIALS AND METHODS

1. Central Data Processing (CTD): operates as a local manager, orchestrating and distributing the data captured in the physical environment. 2. Weather Station: A weather station was built, equipped with an ESP32 LoRa microcontroller, and sensors such as DHT11, Soil Moisture Sensor and hall sensors for the rain gauge. 3. Remote Server: A NodeJs-based server was developed, securely hosted in the cloud, capable of receiving requests over the internet and communicating with a Postgres database. The remote environment has the important responsibility of managing all the data securely and authentically. The entire detailed process is represented in the results and discussions.

ACKNOWLEDGEMENTS

As we conclude this study, it is evident that the collaboration between the Receita Federal, Instituto Federal do Triângulo Mineiro, Universidade Federal de Uberlândia, and Vegetable Breeding Laboratory of the Escola de Agricultura “Luiz de Queiroz” has yielded not only tangible results but also laid a strong foundation for future collaborative initiatives. The intersection of institutional knowledge, technical expertise, and practical experience has provided valuable insights that contribute to the ongoing advancement of technological innovation and social development. The authors would like to express their gratitude to the Receita Federal do Brasil for placing its trust in the educational institutions involved in this project.